

MLOPS

или как построить инфраструктуру для ML

PLAN A

1. Зачем нужен MLOps?
2. Общая схема
3. Инструменты

PLAN B

PLAN A

1. Зачем нужен MLOps?
2. Общая схема
3. Инструменты

PLAN B

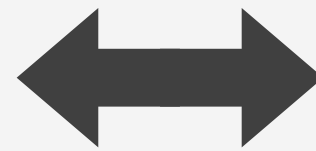
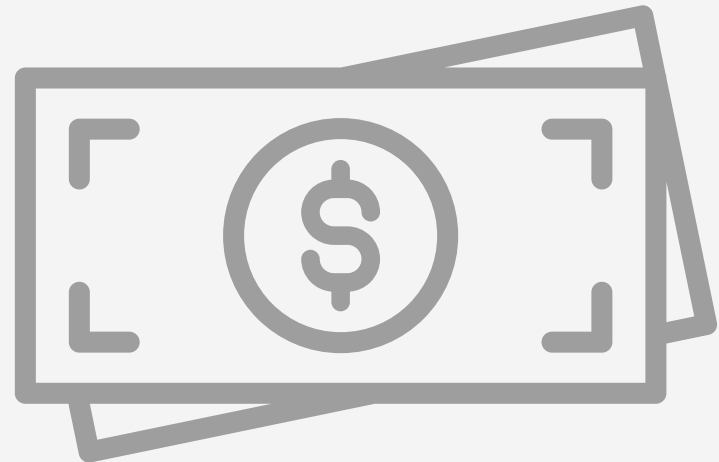


ЦЕЛЬ ВНЕДРЕНИЯ ML

ЦЕЛЬ ВНЕДРЕНИЯ ML



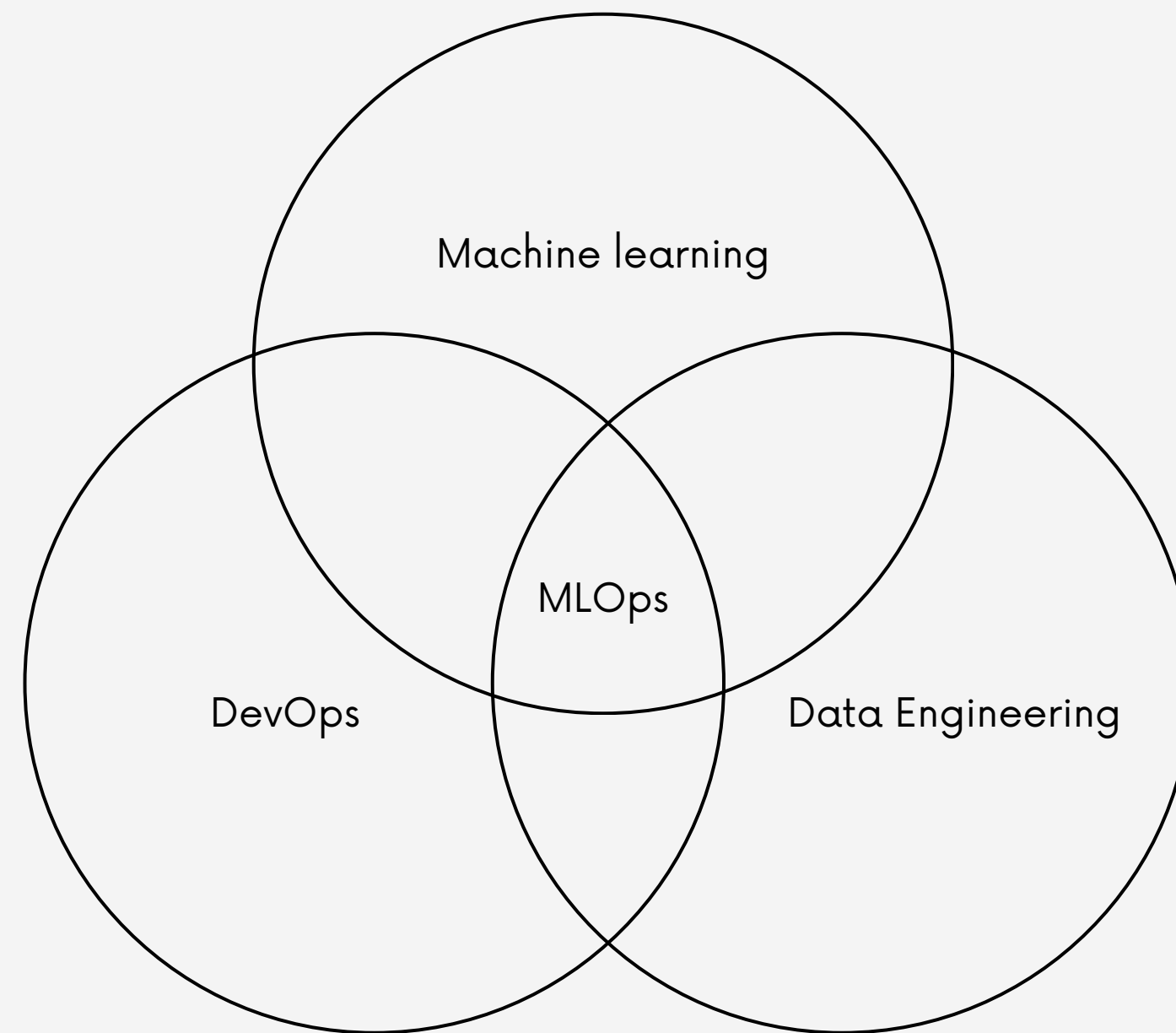
ЦЕЛЬ ВНЕДРЕНИЯ ML



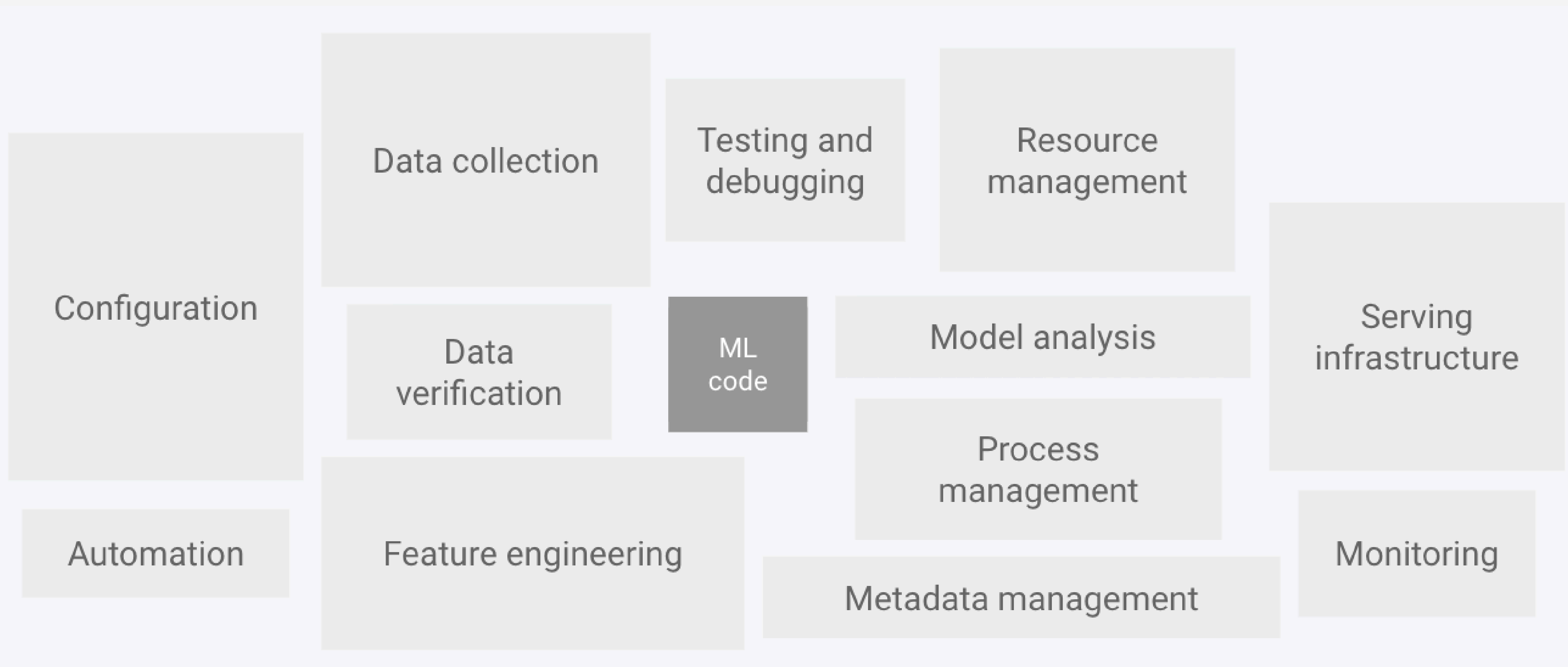
production

MLOPS

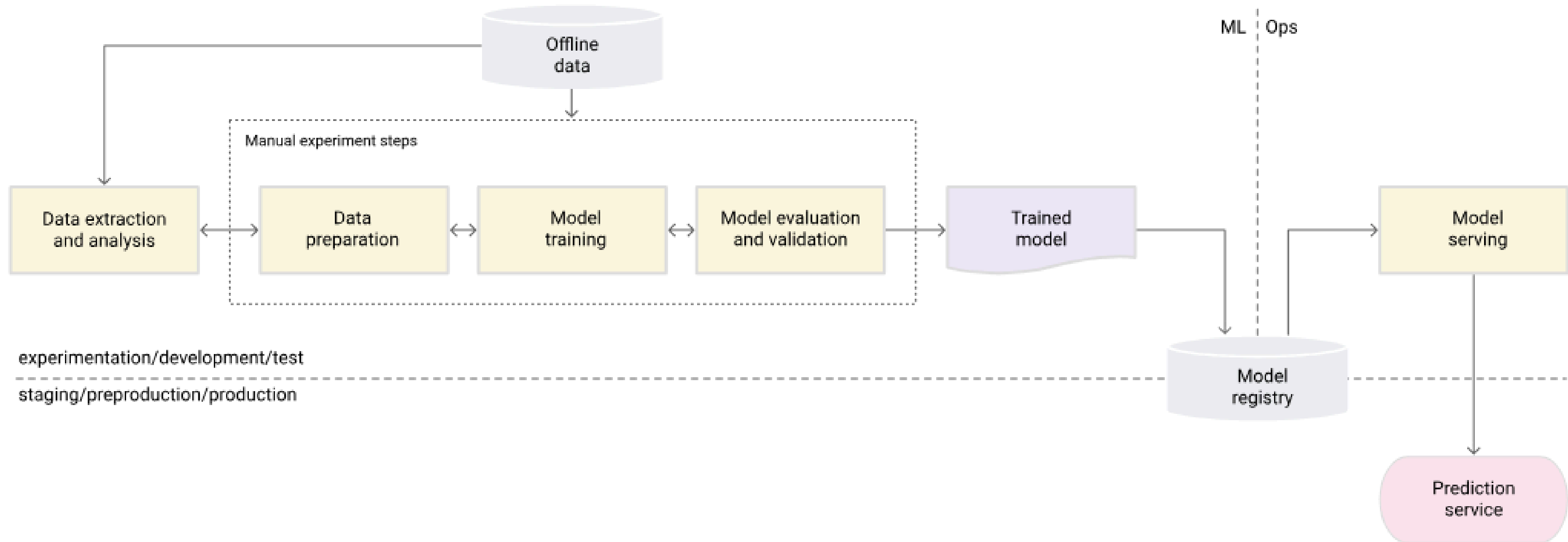
MACHINE LEARNING OPEARATIONS



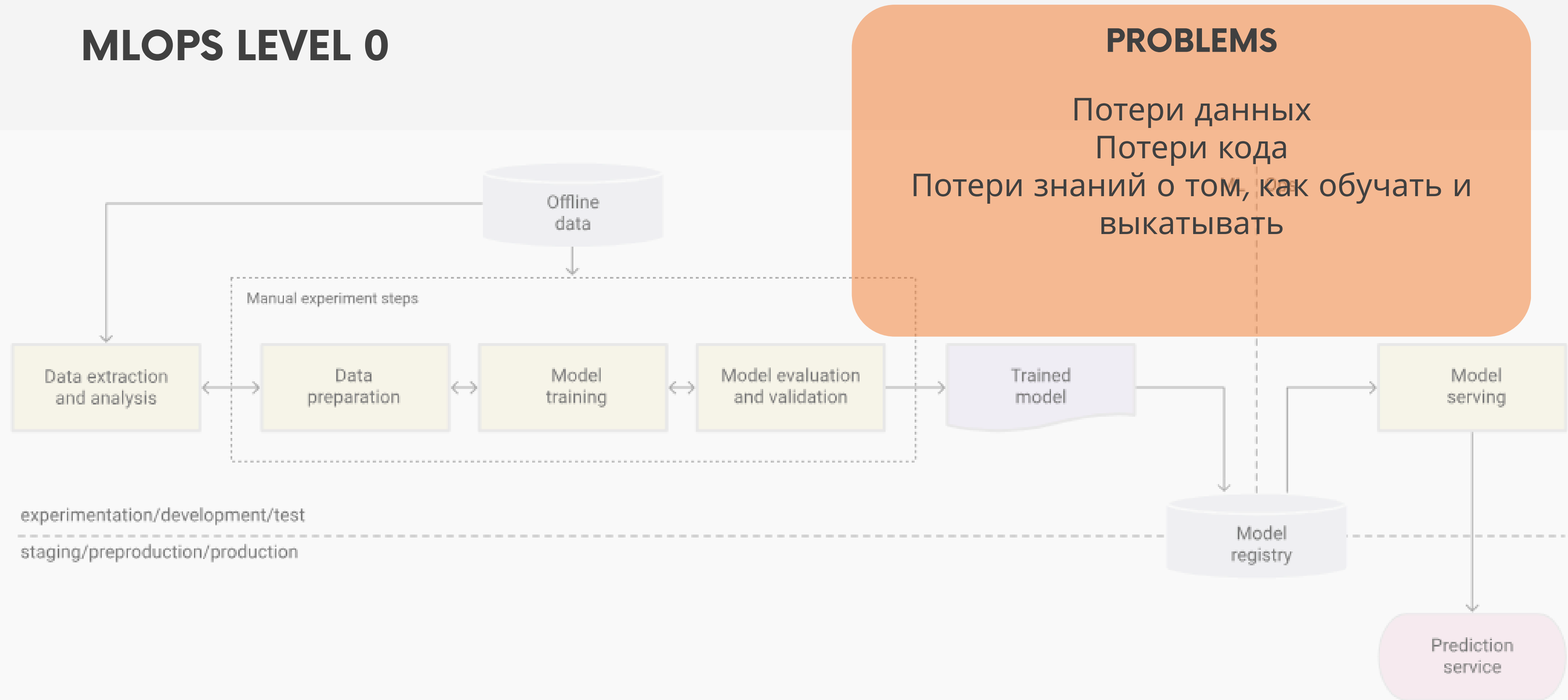
ML-КОД — ЛИШЬ МАЛАЯ ЧАСТЬ РЕАЛЬНОЙ ML-СИСТЕМЫ



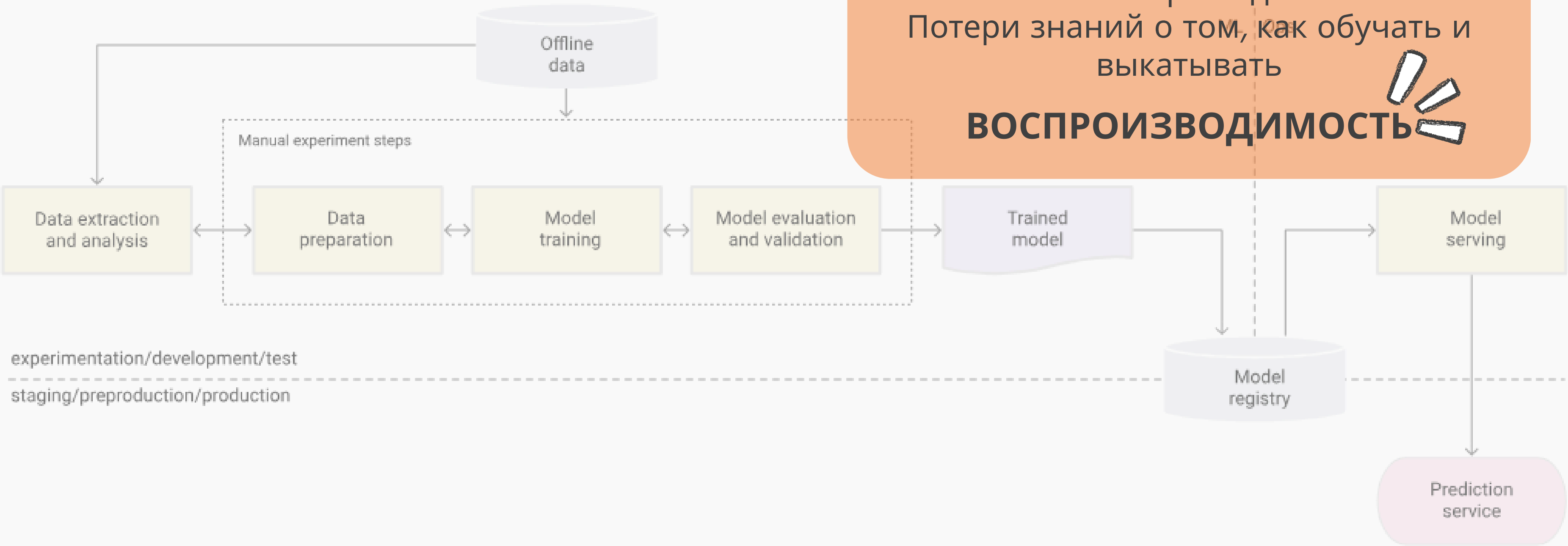
MLOPS LEVEL 0



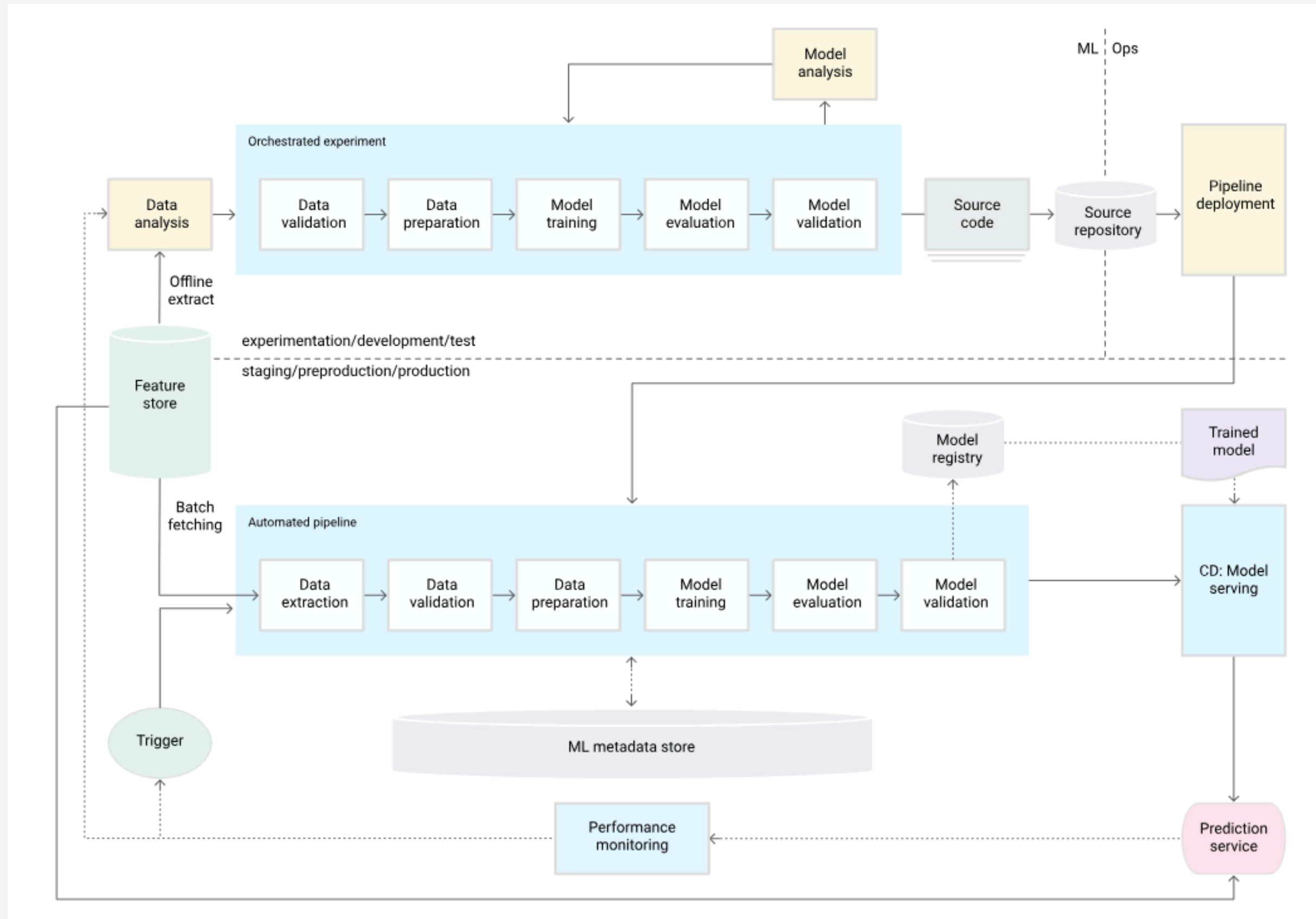
MLOPS LEVEL 0



MLOPS LEVEL 0

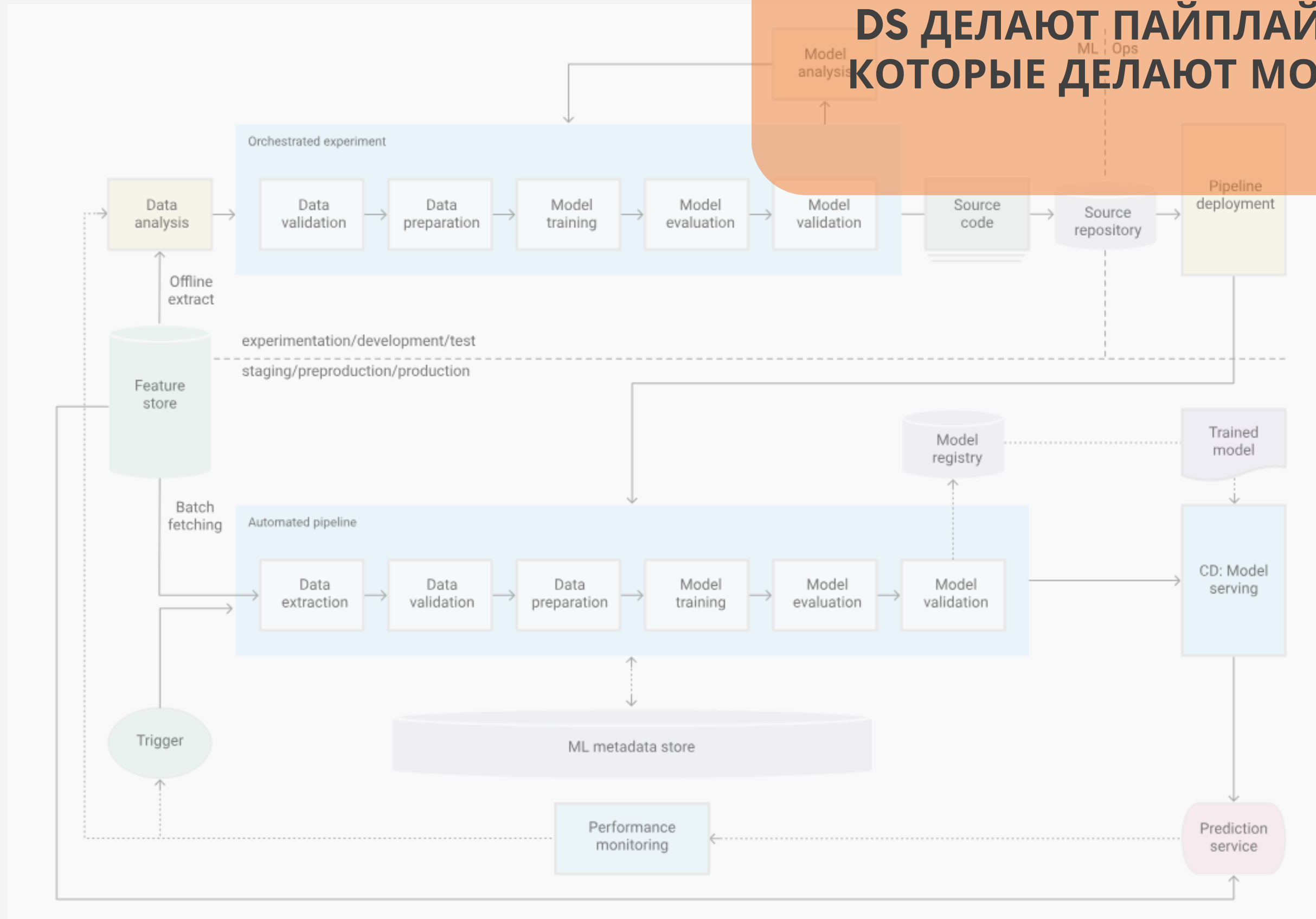


MLOPS NEXT LEVEL

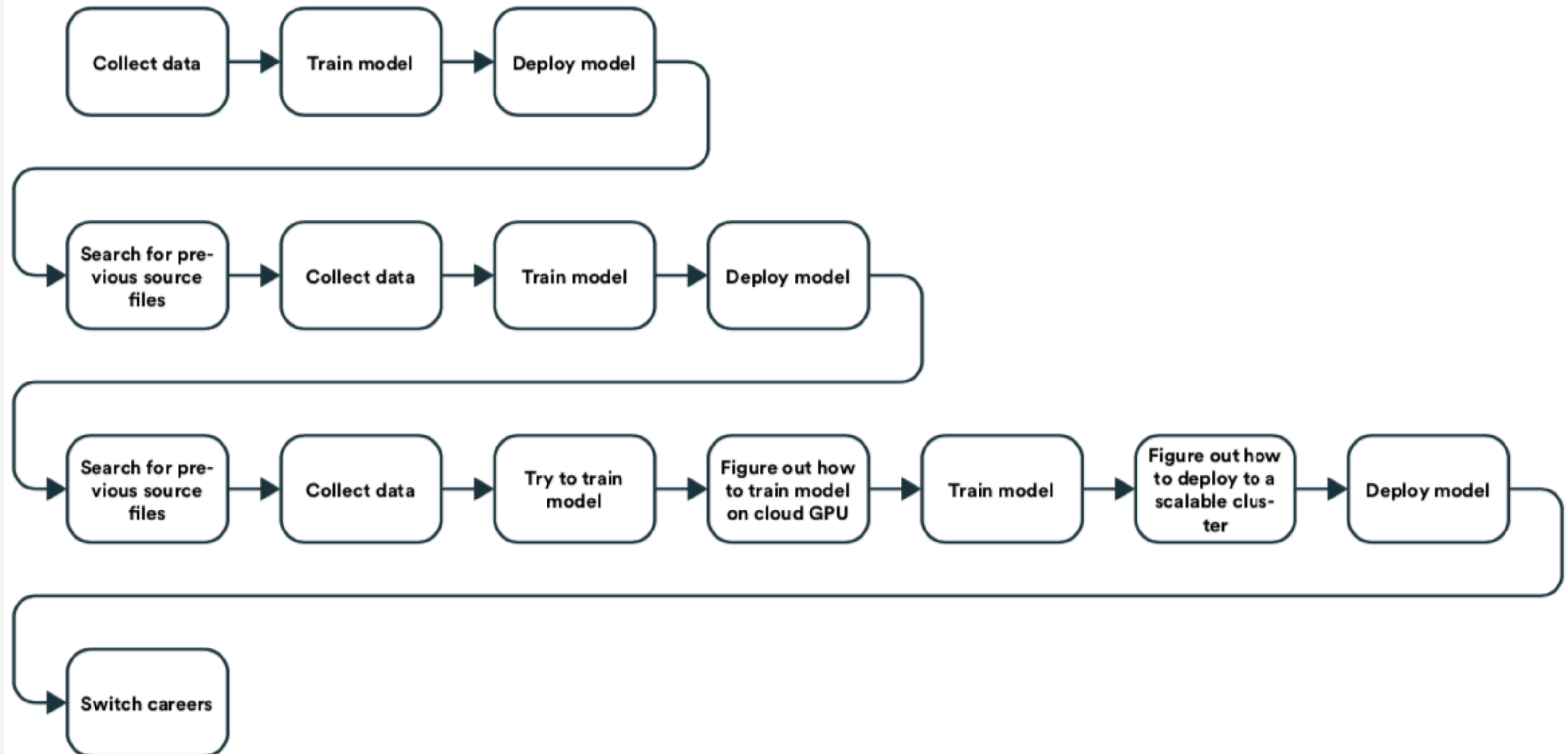


MLOPS NEXT LEVEL

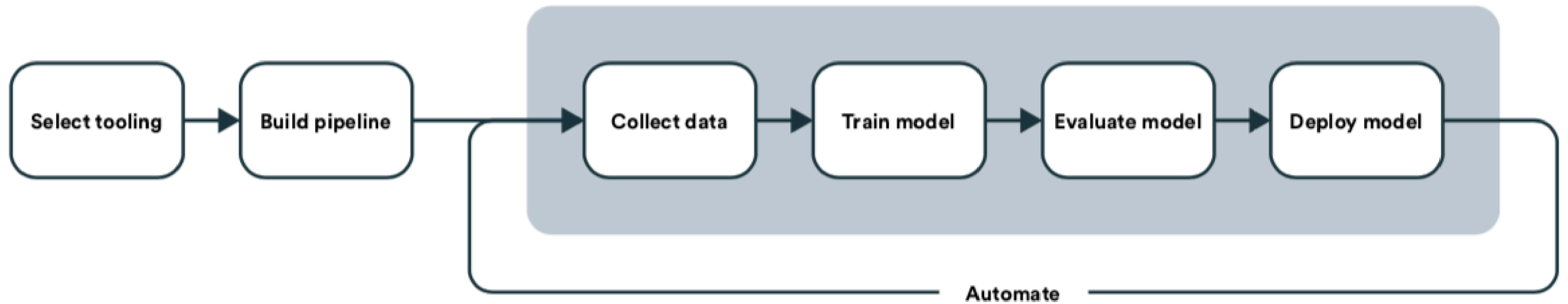
**DS ДЕЛАЮТ ПАЙПЛАЙНЫ,
КОТОРЫЕ ДЕЛАЮТ МОДЕЛИ** !



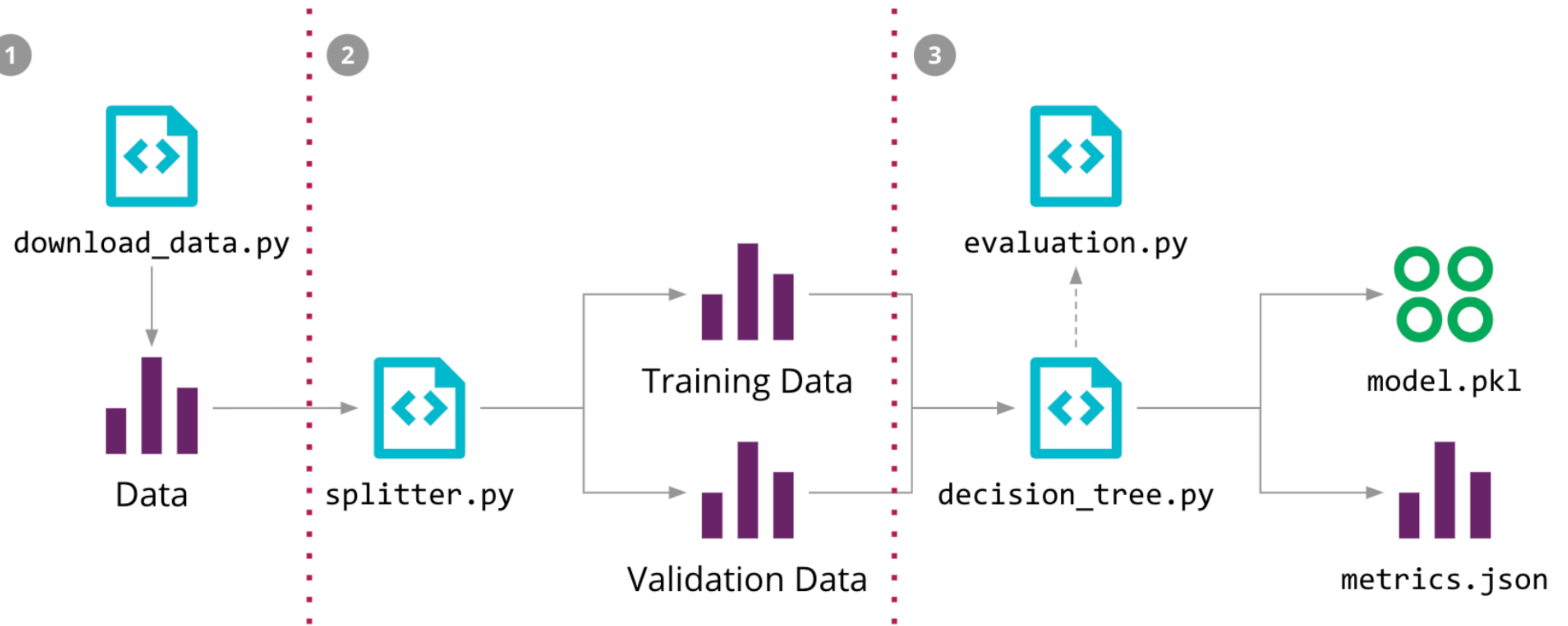
MLOPS LEVEL 0



MLOPS NEXT LEVEL



СТРУКТУРА DS ПРОЕКТА



СТРУКТУРА DS ПРОЕКТА: COOKIECUTTER

LICENSE	<- Open-source license if one is chosen
Makefile	<- Makefile with convenience commands like `make data` or `make train`
README.md	<- The top-level README for developers using this project.
data	
external	<- Data from third party sources.
interim	<- Intermediate data that has been transformed.
processed	<- The final, canonical data sets for modeling.
raw	<- The original, immutable data dump.
docs	<- A default mkdocs project; see www.mkdocs.org for details
models	<- Trained and serialized models, model predictions, or model summaries
notebooks	<- Jupyter notebooks. Naming convention is a number (for ordering), the creator's initials, and a short `-` delimited description, e.g. `1.0-jqp-initial-data-exploration`.
pyproject.toml	<- Project configuration file with package metadata for {{ cookiecutter.module_name }} and configuration for tools like blac
references	<- Data dictionaries, manuals, and all other explanatory materials.
reports	<- Generated analysis as HTML, PDF, LaTeX, etc.
figures	<- Generated graphics and figures to be used in reporting
requirements.txt	<- The requirements file for reproducing the analysis environment, e.g. generated with `pip freeze > requirements.txt`
setup.cfg	<- Configuration file for flake8
{{ cookiecutter.module_name }}	<- Source code for use in this project.
__init__.py	<- Makes {{ cookiecutter.module_name }} a Python module
config.py	<- Store useful variables and configuration
dataset.py	<- Scripts to download or generate data
features.py	<- Code to create features for modeling
modeling	
__init__.py	
predict.py	<- Code to run model inference with trained models
train.py	<- Code to train models
plots.py	<- Code to create visualizations

CODESTYLE



Pylint

Pylint & Flake8 in Python



Flake8



PEP 8

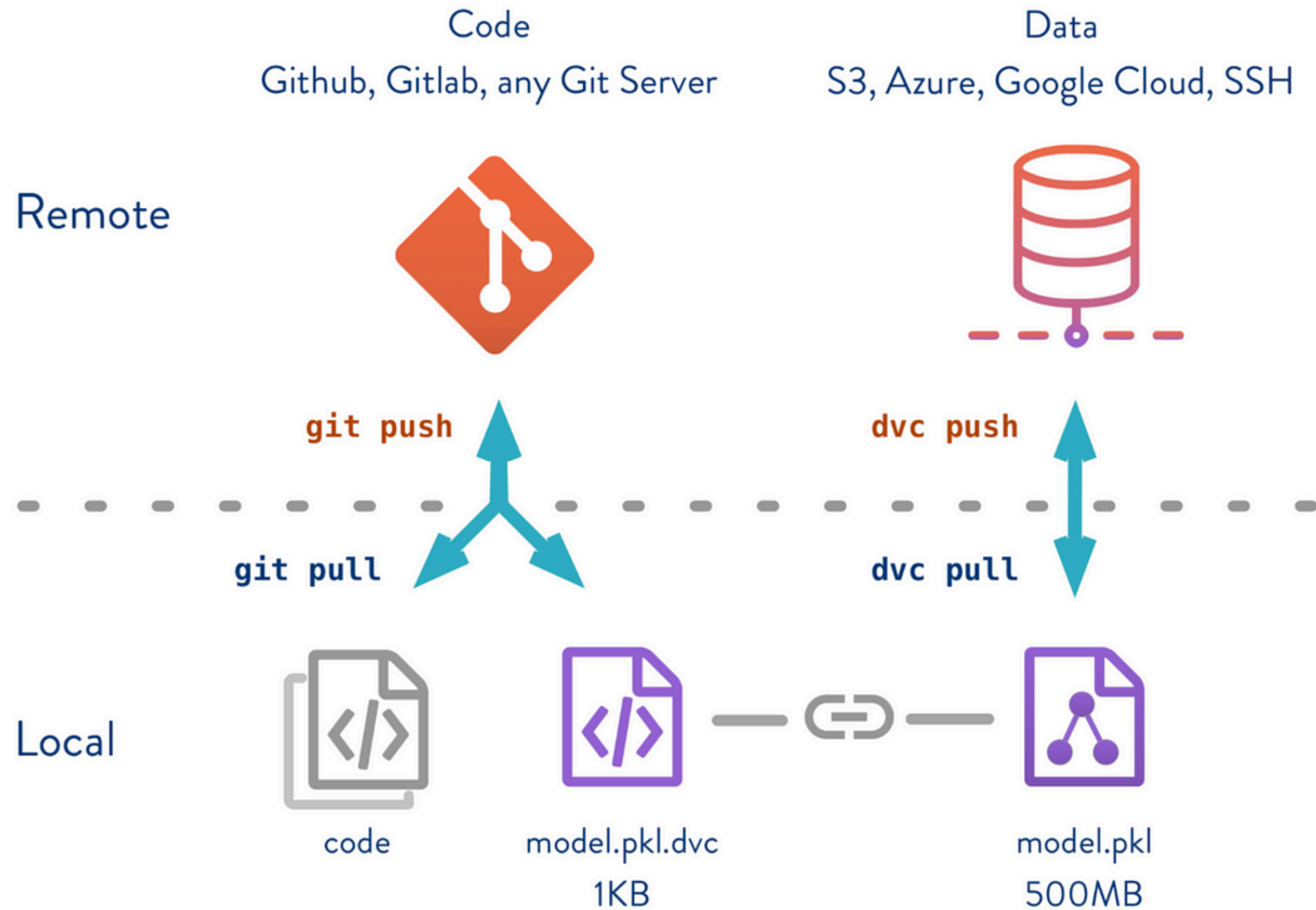


Style Guide

isort

isort your imports, so you don't have to.

DATA VERSION CONTROL



MLFLOW



MLflow is an open source platform to manage the ML lifecycle, including experimentation, reproducibility, deployment, and a central model registry. MLflow currently offers four components:

MLflow Tracking

Record and query experiments: code, data, config, and results

[Read more](#)

MLflow Projects

Package data science code in a format to reproduce runs on any platform

[Read more](#)

MLflow Models

Deploy machine learning models in diverse serving environments

[Read more](#)

Model Registry

Store, annotate, discover, and manage models in a central repository

[Read more](#)

MLFLOW TRACKING



mlflow1.26.1

ExperimentsModels

GitHubDocs

Experiments

+<iris

Search Experiments

Defaultiris

Experiment ID: 1

DescriptionEdit

RefreshCompareDeleteDownload CSV↑ break_tiesAll time

ColumnsOnly show differencesmetrics.rmse < 1 and params.model = "tree"SearchFilterClear

Showing 18 matching runs

	Start Time	Duration	Run Name	User	Source	Version	Models	training_accurac	training_f1_scor	training_precisic	C
☐	3 days ago	8.7s	-	Igor	D:\Works...	7762f0	sklearn	0.983	0.983	0.984	1.0
☐	2 days ago	14.6s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	1
☐	2 days ago	13.4s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	1
☐	2 days ago	7.4s	-	Igor	D:\Works...	7762f0	-	-	-	-	1
☐	2 days ago	9.8s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	1
☐	2 days ago	8.6s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	1
☐	2 days ago	8.4s	-	Igor	D:\Works...	7762f0	sklearn	0.775	0.773	0.777	2
☐	2 days ago	8.3s	-	Igor	D:\Works...	7762f0	sklearn	0.775	0.773	0.777	100
☐	2 days ago	8.8s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	100
☐	2 days ago	8.5s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	100
☐	2 days ago	8.5s	-	Igor	D:\Works...	7762f0	sklearn	0.783	0.782	0.785	100
☐	2 days ago	8.7s	-	Igor	D:\Works...	7762f0	sklearn	0.867	0.867	0.867	100

Registered Models > Iris

Iris

Created Time: 2022-06-20 11:21:18

Last Modified: 2022-06-20 11:25:57

▼ Description [Edit](#)

Модель классификации цветков ириса

► Tags

▼ Versions

AllActive 2Compare

☐	Version	Registered at	Created by	Stage	Description
☐	✔ Version 2	2022-06-20 11:21:54		Staging	Модель релиз кандидат
☐	✔ Version 1	2022-06-20 11:21:37		Production	Модель используемая в Production

ОРЕКСТРАТОР

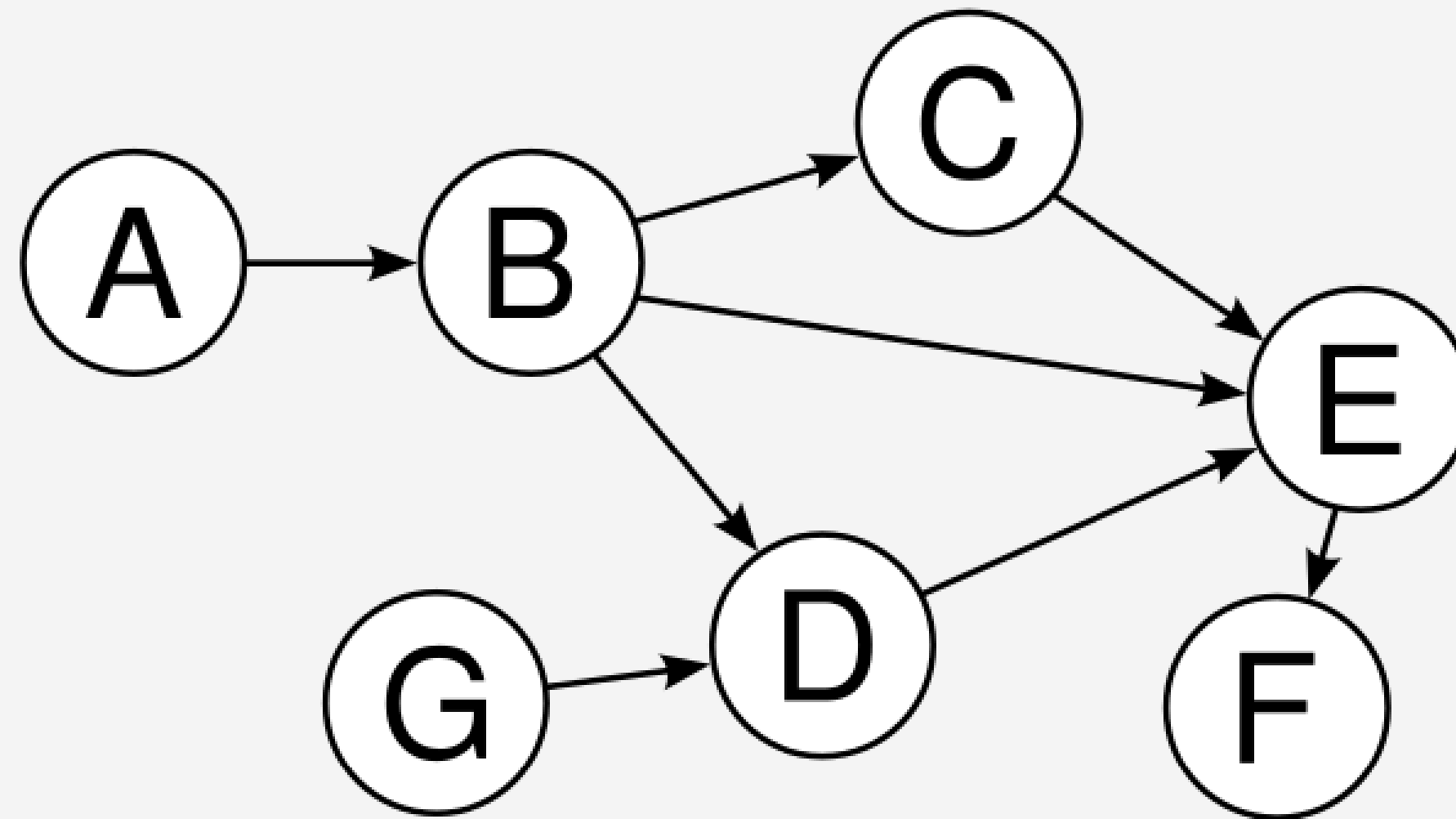


- **Планирование задач** – когда запустить
- **Управление зависимостями** – ждать, пока исполнятся другие задачи перед запуском)
- **Репроцессинг** – легко перезапускать упавшие задачи и зависящие от неё
- **Мониторинг** – если задача упала, то нужно об этом уведомить

DAG



Apache
Airflow



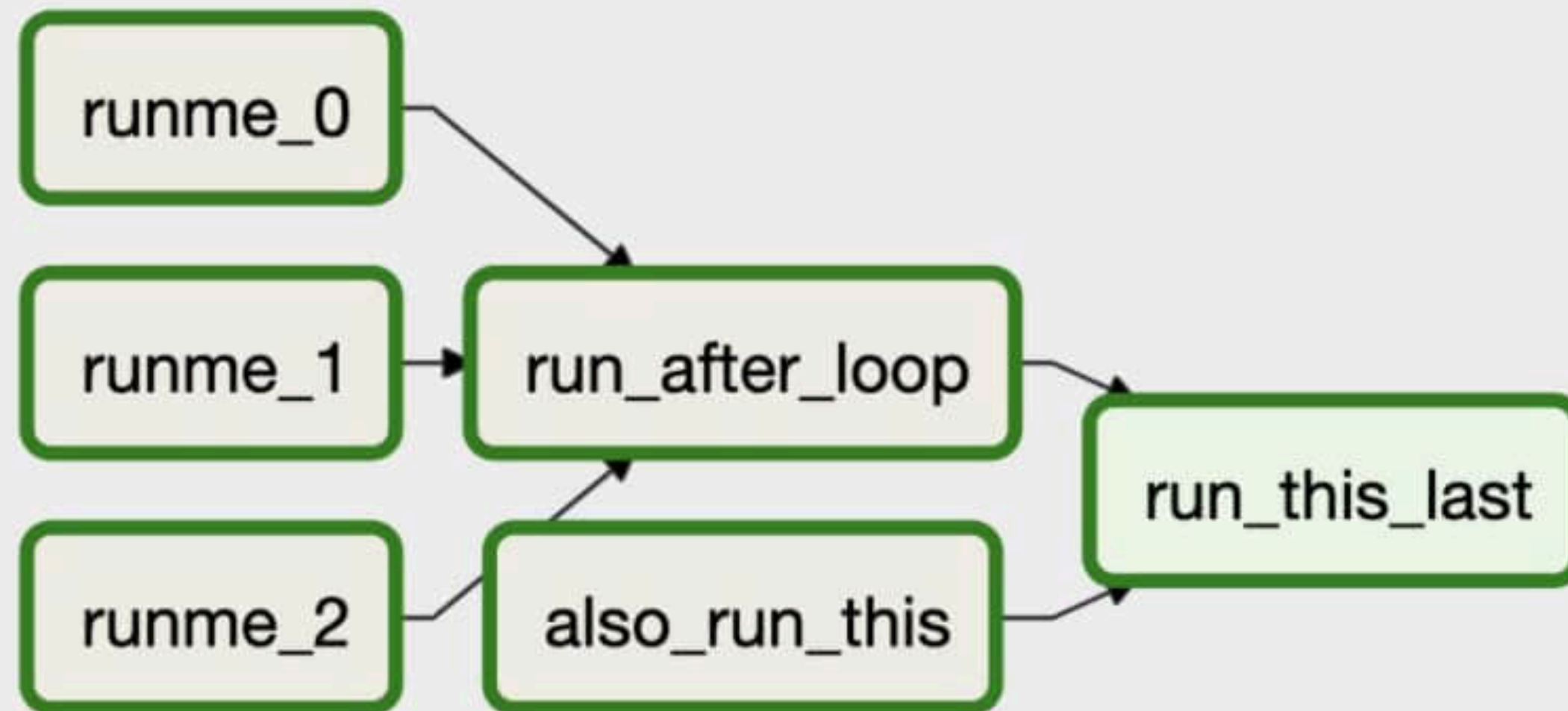
AIRFLOW DAGS



DAGs

All 26Active 3Paused 23			Filter DAGs by tag		Search DAGs		
DAG	Owner	Runs	Schedule	Last Run	Recent Tasks	Actions	Links
example_bash_operator exampleexample2	airflow	2	00***	2021-05-04, 00:00:00	6		
example_branch_dop_operator_v3 example	airflow	919 1	*/* ****	2021-05-03, 15:19:00	4 2		
example_branch_operator exampleexample2	airflow		@daily				
example_complex exampleexample2example3	airflow		None				
example_dag_decorator example	airflow		None				

AIRFLOW TASKS



FASTAPI



GET

/ Read Data

POST

/ Write Data

GET

/ {id} Read Data

PUT

/ {id} Update Data

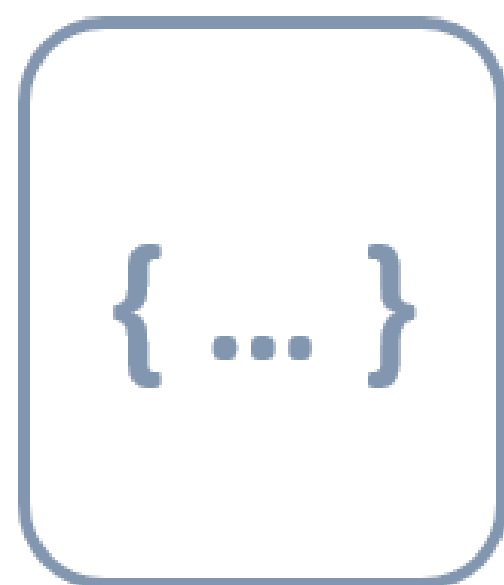
DELETE

/ {id} Delete Data

DOCKER



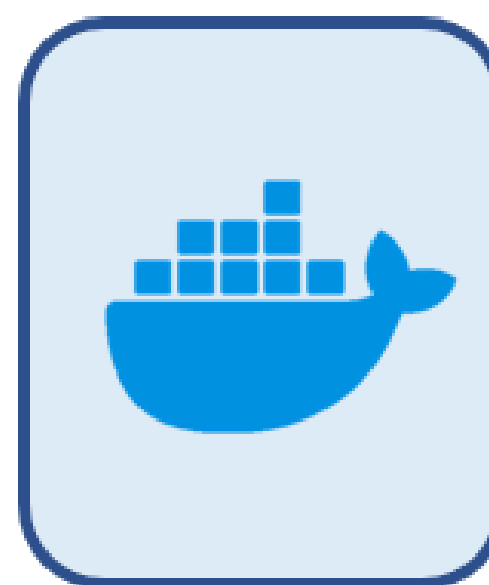
Docker File



Build



Docker Image



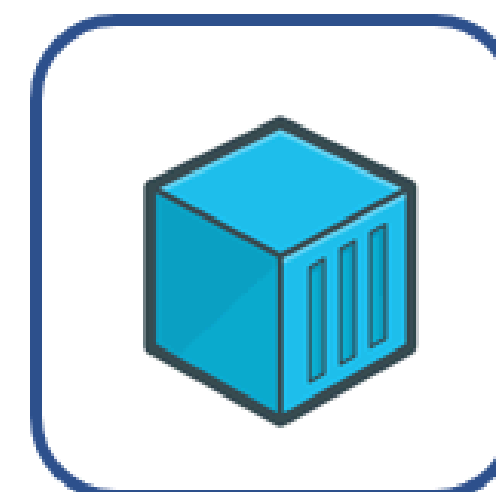
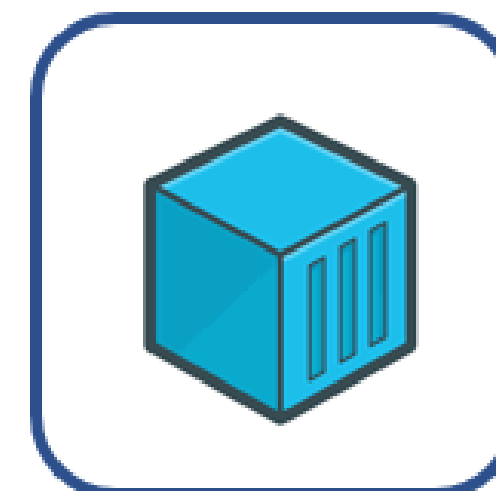
Run



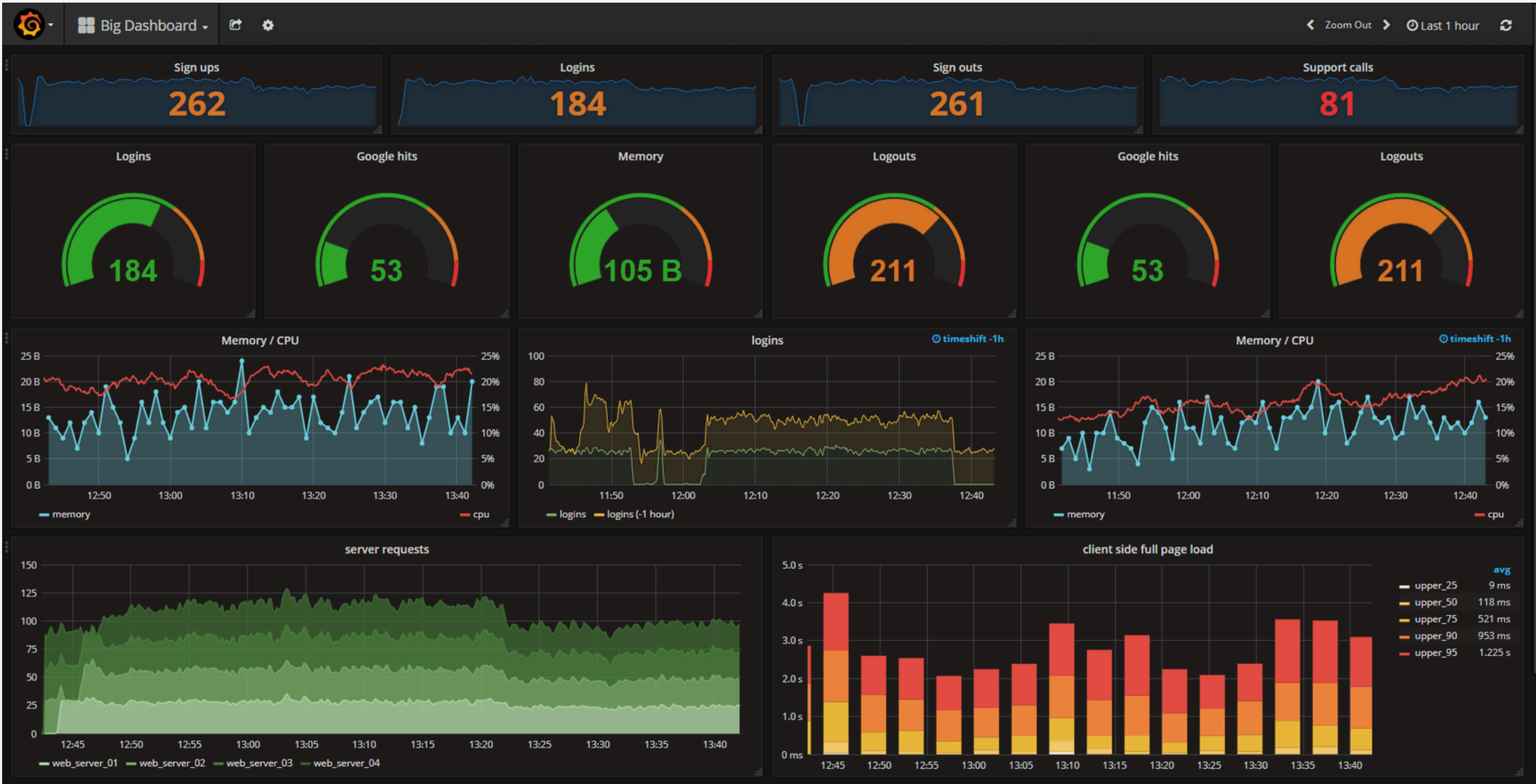
Run



Docker Container



GRAFANA



CONTACTS & QUESTIONS

